Contents

Chapter I

Curves

§1 Parametrized Curves

Definition 1.1. *A* ***parametrized differentiable curve*** *is a differentiable map*

$$\alpha : I \to \mathbb{R}^3$$

from an open interval $I = (a, b) \subset \mathbb{R}$ *into* $\mathbb{R}^3$*. If* $\alpha(t) = (x(t), y(t), z(t))$*, differentiability means that* x, y, z *are differentiable functions.*

Definition 1.2. *The vector*

$$\alpha'(t) = (x'(t), y'(t), z'(t))$$

is called the ***tangent vector****, or* ***velocity vector****, of* α *at* t*. The image set* $\alpha(I) \subset \mathbb{R}^3$ *is called the* ***trace*** *of* α*.*

Example 1.3. *The curve*

$$\alpha(t) = (a \cos t, a \sin t, bt), \qquad t \in \mathbb{R},$$

has as its trace a helix of pitch $2\pi b$ *on the cylinder* $x^2 + y^2 = a^2$*.*

Definition 1.4. *A parametrized differentiable curve* $\alpha : I \to \mathbb{R}^3$ *is said to be* ***regular*** *if*

$$\alpha'(t) \neq 0$$

for every $t \in I$*.*

Definition 1.5. *Let* $\alpha : I \to \mathbb{R}^3$ *be a regular parametrized curve. The* ***arc length*** *of* α *from* $t_0 \in I$ *to* $t \in I$ *is*

$$s(t) = \int_{t_0}^{t} |\alpha'(u)| \, du.$$

If $|\alpha'(s)| = 1$*, the curve is said to be* ***parametrized by arc length****.*

Remark. *Since $ds/dt = |\alpha'(t)| > 0$, the arc length function has a differentiable inverse locally. Thus every regular curve may be reparametrized by arc length.*

§2 The Local Theory of Curves Parametrized by Arc Length

Definition 2.1. *Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length s. The number*

$$\kappa(s) := |\alpha''(s)|$$

*is called the **curvature** of α at s.*

Definition 2.2. *Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length, with $\alpha''(s) \neq 0$ for all $s \in I$. Define*

$$\mathbf{t}(s) := \alpha'(s), \qquad \mathbf{n}(s) := \frac{\alpha''(s)}{\kappa(s)}, \qquad \mathbf{b}(s) := \mathbf{t}(s) \times \mathbf{n}(s).$$

*These are called the **unit tangent vector**, **principal normal vector**, and **binormal vector**, respectively.*

Definition 2.3. *The **torsion** $\tau(s)$ is defined by*

$$\mathbf{b}'(s) = \tau(s)\mathbf{n}(s).$$

Definition 2.4. *Let α be as above.*

1. *The plane determined by $\mathbf{t}(s)$ and $\mathbf{n}(s)$ is called the **osculating plane** at s.*
2. *The plane determined by $\mathbf{n}(s)$ and $\mathbf{b}(s)$ is called the **normal plane** at s.*
3. *The plane determined by $\mathbf{t}(s)$ and $\mathbf{b}(s)$ is called the **rectifying plane** at s.*

Theorem 2.5 (Frenet Formulas)**.** *Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length and assume $\alpha''(s) \neq 0$ for all $s \in I$. Then*

$$\begin{aligned} \mathbf{t}' &= \kappa\mathbf{n}, \\ \mathbf{n}' &= -\kappa\mathbf{t} - \tau\mathbf{b}, \\ \mathbf{b}' &= \tau\mathbf{n}. \end{aligned}$$

Equivalently,

$$\begin{pmatrix} \mathbf{t}' \\ \mathbf{n}' \\ \mathbf{b}' \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & -\tau \\ 0 & \tau & 0 \end{pmatrix} \begin{pmatrix} \mathbf{t} \\ \mathbf{n} \\ \mathbf{b} \end{pmatrix}.$$

Theorem 2.6 (Fundamental Theorem of the Local Theory of Curves)**.** *Given differentiable functions $\kappa(s) > 0$ and $\tau(s)$ on an interval I, there exists a regular parametrized curve $\alpha : I \to \mathbb{R}^3$ such that s is arc length, $\kappa(s)$ is its curvature, and $\tau(s)$ is its torsion. Moreover, any other such curve differs from α by a rigid motion.*

§3 The Local Canonical Form

Remark. *This section expresses a curve near a point in the coordinates determined by its Frenet trihedron. It gives a local Taylor expansion whose coefficients are written in terms of curvature and torsion.*

§4 Global Properties of Plane Curves

Remark. *This section studies global restrictions on plane curves, including results such as the isoperimetric inequality and the four-vertex theorem.*

Chapter II

Regular Surfaces

§1 Introduction

The central object of this chapter is a regular surface in $\mathbb{R}^3$. The goal is to develop a workable local model for surfaces using parametrizations, tangent planes, differentials, and the first fundamental form. The guiding idea is that a surface is a set which can be described locally by two independent parameters.

§2 Regular Surfaces; Inverse Images of Regular Values

Definition 2.1. *A subset $S \subset \mathbb{R}^3$ is a **regular surface** if, for each $p \in S$, there exist an open set $V \subset \mathbb{R}^3$ containing p, an open set $U \subset \mathbb{R}^2$, and a map $\mathbf{x} : U \to V \cap S$ such that*

(i) $\mathbf{x}$ *is differentiable;*

(ii) $\mathbf{x}$ *is a homeomorphism;*

(iii) for each $q \in U$, the differential $d\mathbf{x}_q$ is injective.

Such a map $\mathbf{x}$ *is called a **parametrization** of S.*

Proposition 2.2. *If $f : U \to \mathbb{R}$ is a differentiable function on an open set $U \subset \mathbb{R}^2$, then the graph*

$$\{(x, y, f(x, y)) : (x, y) \in U\}$$

is a regular surface.

Proposition 2.3. *If $f : U \subset \mathbb{R}^3 \to \mathbb{R}$ is differentiable and $a \in f(U)$ is a regular value of f, then $f^{-1}(a)$ is a regular surface in $\mathbb{R}^3$.*

Example 2.4. *The unit sphere*

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$$

is a regular surface, since it is the inverse image of the regular value 1 *of* $f(x, y, z) = x^2 + y^2 + z^2$.

Example 2.5. *A torus with radii* $a > r > 0$ *can be written as the inverse image of* r^2 *by*

$$f(x, y, z) = z^2 + \left(\sqrt{x^2 + y^2} - a\right)^2.$$

A useful parametrization is

$$\mathbf{x}(u, v) = ((a + r\cos u)\cos v, (a + r\cos u)\sin v, r\sin u),$$

where the parameters are taken in suitable open intervals.

Proposition 2.6. *Let* $S \subset \mathbb{R}^3$ *be a regular surface and let* $p \in S$. *Then some neighborhood of* p *in* S *is the graph of a differentiable function of one of the following forms:*

$$z = f(x, y), \qquad y = g(x, z), \qquad x = h(y, z).$$

Proof. Let $\mathbf{x}(u, v) = (x(u, v), y(u, v), z(u, v))$ be a parametrization at p, and set $q = \mathbf{x}^{-1}(p)$. Since $d\mathbf{x}_q$ is injective, at least one of

$$\frac{\partial(x, y)}{\partial(u, v)}, \qquad \frac{\partial(y, z)}{\partial(u, v)}, \qquad \frac{\partial(z, x)}{\partial(u, v)}$$

is nonzero at q. If, for instance, $\partial(x, y)/\partial(u, v) \neq 0$, the inverse function theorem applies to $(u, v) \mapsto (x(u, v), y(u, v))$. Locally one may solve $(u, v) = (u(x, y), v(x, y))$, and the surface is given by $z = z(u(x, y), v(x, y))$. The other cases are the same after changing the projected coordinates. □

Proposition 2.7. *Let* S *be a regular surface and let* $\mathbf{x} : U \subset \mathbb{R}^2 \to \mathbb{R}^3$ *satisfy* $\mathbf{x}(U) \subset S$. *If* $\mathbf{x}$ *is differentiable, one-to-one, and* $d\mathbf{x}_q$ *is injective for all* $q \in U$, *then* $\mathbf{x}^{-1}$ *is continuous. Hence* $\mathbf{x}$ *is a parametrization of its image.*

Remark. *The cone* $z = \sqrt{x^2 + y^2}$ *is not a regular surface at the origin. If it were regular there, it would locally be the graph of a differentiable function over one of the coordinate planes; the only possible graph over the* xy*-plane is not differentiable at* $(0, 0)$.

§3 Change of Parameters; Differentiable Functions on Surfaces

Proposition 3.1 (Change of Parameters)**.** *Let* $\mathbf{x} : U \subset \mathbb{R}^2 \to S$ *and* $\mathbf{y} : V \subset \mathbb{R}^2 \to S$ *be parametrizations of a regular surface* S. *If*

$$W = \mathbf{x}(U) \cap \mathbf{y}(V) \neq \varnothing,$$

then the change of coordinates

$$h = \mathbf{x}^{-1} \circ \mathbf{y} : \mathbf{y}^{-1}(W) \to \mathbf{x}^{-1}(W)$$

is a diffeomorphism.

Proof. Continuity follows from the fact that parametrizations are homeomorphisms. For differentiability, fix a point in the overlap and choose coordinates so that $\partial(x, y)/\partial(u, v) \neq 0$ for the parametrization $\mathbf{x}(u, v) = (x(u, v), y(u, v), z(u, v))$. Extend $\mathbf{x}$ to

$$F(u, v, t) = (x(u, v), y(u, v), z(u, v) + t).$$

The differential of F has nonzero determinant at the point, so F has a local differentiable inverse. Locally, $h = F^{-1} \circ \mathbf{y}$ followed by projection to the first two coordinates. The same argument applied to h^{-1} proves that h is a diffeomorphism. $\square$

Definition 3.2. *Let $V \subset S$ be open and let $f : V \to \mathbb{R}$. We say that f is* ***differentiable at*** *$p \in V$ if, for some parametrization $\mathbf{x} : U \to S$ with $p \in \mathbf{x}(U)$, the function*

$$f \circ \mathbf{x} : U \to \mathbb{R}$$

is differentiable at $\mathbf{x}^{-1}(p)$.

Remark. *The change of parameters proposition shows that the definition does not depend on the parametrization. If two coordinate systems are related by a diffeomorphism, differentiability in one system is equivalent to differentiability in the other.*

Definition 3.3. *A map $\varphi : S_1 \to S_2$ between regular surfaces is* ***differentiable at*** *$p \in S_1$ if, for parametrizations $\mathbf{x} : U \to S_1$ and $\mathbf{y} : V \to S_2$ with $p \in \mathbf{x}(U)$ and $\varphi(p) \in \mathbf{y}(V)$, the local expression*

$$\mathbf{y}^{-1} \circ \varphi \circ \mathbf{x}$$

is differentiable near $\mathbf{x}^{-1}(p)$.

Definition 3.4. *A differentiable map $\varphi : S_1 \to S_2$ is a* ***diffeomorphism*** *if it is one-to-one, onto, and its inverse is also differentiable. In this case S_1 and S_2 are* ***diffeomorphic****.*

§4 The Tangent Plane; The Differential of a Map

Definition 4.1. *Let S be a regular surface and $p \in S$. The* ***tangent plane*** *T_pS is the set of tangent vectors at p of differentiable curves on S passing through p.*

Proposition 4.2. *If $\mathbf{x} : U \to S$ is a parametrization and $p = \mathbf{x}(u_0, v_0)$, then*

$$T_pS = d\mathbf{x}_{(u_0,v_0)}(\mathbb{R}^2) = \text{span}\{\mathbf{x}_u(u_0, v_0), \mathbf{x}_v(u_0, v_0)\}.$$

In particular $\{\mathbf{x}_u, \mathbf{x}_v\}$ is a basis of T_pS.

Definition 4.3. *Let $\varphi : S_1 \to S_2$ be differentiable and let $p \in S_1$. The* ***differential*** *of φ at p is the linear map*

$$d\varphi_p : T_pS_1 \to T_{\varphi(p)}S_2$$

defined as follows: if $w = \alpha'(0)$ for a curve $\alpha : (-\epsilon, \epsilon) \to S_1$ with $\alpha(0) = p$, then

$$d\varphi_p(w) = (\varphi \circ \alpha)'(0).$$

Remark. *If $\mathbf{x}$ and $\mathbf{y}$ are local parametrizations of S_1 and S_2, and*

$$\Phi = \mathbf{y}^{-1} \circ \varphi \circ \mathbf{x},$$

then $d\varphi_p$ is represented in the bases $\{\mathbf{x}_u, \mathbf{x}_v\}$ and $\{\mathbf{y}_\xi, \mathbf{y}_\eta\}$ by the Jacobian matrix of Φ.

Definition 4.4. *A differentiable map $\varphi : S_1 \to S_2$ is a* ***local diffeomorphism at*** *p if there are neighborhoods U_1 of p and U_2 of $\varphi(p)$ such that $\varphi : U_1 \to U_2$ is a diffeomorphism.*

Proposition 4.5. *If $\varphi : S_1 \to S_2$ is differentiable and $d\varphi_p : T_pS_1 \to T_{\varphi(p)}S_2$ is an isomorphism, then φ is a local diffeomorphism at p.*

§5 The First Fundamental Form; Area

Definition 5.1. *Let S be a regular surface in $\mathbb{R}^3$ and $p \in S$. The natural inner product of $\mathbb{R}^3$ induces on T_pS an inner product $\langle , \rangle_p$. The corresponding quadratic form*

$$I_p(w) = \langle w, w \rangle_p = |w|^2, \qquad w \in T_pS,$$

is called the ***first fundamental form*** *of S at p.*

Remark. *If $\mathbf{x}(u, v)$ is a parametrization and $\alpha(t) = \mathbf{x}(u(t), v(t))$, then*

$$\begin{aligned} I_p(\alpha'(0)) &= \langle \mathbf{x}_u u' + \mathbf{x}_v v', \mathbf{x}_u u' + \mathbf{x}_v v' \rangle_p \\ &= E(u')^2 + 2Fu'v' + G(v')^2, \end{aligned}$$

where

$$E = \langle \mathbf{x}_u, \mathbf{x}_u \rangle, \qquad F = \langle \mathbf{x}_u, \mathbf{x}_v \rangle, \qquad G = \langle \mathbf{x}_v, \mathbf{x}_v \rangle.$$

Definition 5.2. *Let $w_1 = a\mathbf{x}_u + b\mathbf{x}_v$ and $w_2 = c\mathbf{x}_u + d\mathbf{x}_v$. Their inner product is*

$$\langle w_1, w_2 \rangle = Eac + F(ad + bc) + Gbd.$$

Thus the first fundamental form is locally represented by the matrix

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix}.$$

Proposition 5.3. *Let $\alpha(t) = \mathbf{x}(u(t), v(t))$ be a regular curve on S. Its arc length between $t = a$ and $t = b$ is*

$$L(\alpha) = \int_a^b \sqrt{E(u')^2 + 2Fu'v' + G(v')^2}\, dt.$$

Proposition 5.4. *If two regular curves on S meet at p with tangent vectors $w_1, w_2 \in T_pS$, then their angle θ is determined by*

$$\cos\theta = \frac{\langle w_1, w_2\rangle_p}{\sqrt{\langle w_1, w_1\rangle_p}\sqrt{\langle w_2, w_2\rangle_p}}.$$

In particular, the coordinate curves of $\mathbf{x}$ *are orthogonal exactly when $F = 0$.*

Definition 5.5. *A bounded region $R \subset S$ contained in a coordinate neighborhood $\mathbf{x}(U)$ has* ***area***

$$A(R) = \iint_{\mathbf{x}^{-1}(R)} |\mathbf{x}_u \times \mathbf{x}_v|\, du\, dv = \iint_{\mathbf{x}^{-1}(R)} \sqrt{EG - F^2}\, du\, dv.$$

Example 5.6. *For a graph $\mathbf{x}(u, v) = (u, v, f(u, v))$ one has*

$$E = 1 + f_u^2, \qquad F = f_u f_v, \qquad G = 1 + f_v^2,$$

hence

$$dA = \sqrt{1 + f_u^2 + f_v^2}\, du\, dv.$$

Example 5.7. *For the sphere of radius r parametrized by*

$$\mathbf{x}(u, v) = (r \sin u \cos v, r \sin u \sin v, r \cos u),$$

where $0 < u < \pi$, one obtains

$$E = r^2, \qquad F = 0, \qquad G = r^2 \sin^2 u,$$

and therefore $dA = r^2 \sin u\, du\, dv$.

§6 Orientation of Surfaces

Definition 6.1. *A regular surface S is* ***orientable*** *if it can be covered by coordinate neighborhoods in such a way that every change of coordinates between overlapping charts has positive Jacobian. Such a choice of charts is an* ***orientation*** *of S.*

Definition 6.2. *For a parametrization* $\mathbf{x}(u, v)$*, the associated unit normal field is*

$$N = \frac{\mathbf{x}_u \times \mathbf{x}_v}{|\mathbf{x}_u \times \mathbf{x}_v|}.$$

Proposition 6.3. *A regular surface* $S \subset \mathbb{R}^3$ *is orientable if and only if there exists a differentiable field of unit normal vectors* $N : S \to \mathbb{R}^3$.

Proof. If the coordinate changes have positive Jacobian, the local normals $\mathbf{x}_u \times \mathbf{x}_v / |\mathbf{x}_u \times \mathbf{x}_v|$ agree on overlaps and define a global unit normal field. Conversely, if a global unit normal field N is given, choose the order of the parameters in each connected coordinate neighborhood so that the local normal equals N. On an overlap, a negative Jacobian would reverse the local normal, contradicting the choice. □

Example 6.4. *Graphs and level surfaces* $f^{-1}(a)$*, where* a *is a regular value, are orientable. For a level surface one may take*

$$N = \frac{\nabla f}{|\nabla f|}.$$

Example 6.5. *The Mobius strip is nonorientable. When a unit normal is transported once around its central circle, it returns with the opposite sign, so no global continuous choice of unit normal can exist.*

§7 A Characterization of Compact Orientable Surfaces

Definition 7.1. *A regular surface* $S \subset \mathbb{R}^3$ *is* ***compact*** *if it is compact as a subset of* $\mathbb{R}^3$.

Definition 7.2. *Let* $S \subset \mathbb{R}^3$ *be a compact connected regular surface. A point* $p \in \mathbb{R}^3 \setminus S$ *is said to lie in an* ***inside component*** *if it belongs to a bounded connected component of* $\mathbb{R}^3 \setminus S$*. The unbounded component is called the* ***outside****.*

Theorem 7.3 (Jordan-Brouwer Separation for Regular Surfaces)**.** *A compact connected regular surface* $S \subset \mathbb{R}^3$ *separates* $\mathbb{R}^3$ *into exactly two connected components: one bounded and one unbounded. The surface* S *is the common boundary of both components.*

Remark. *In this book the separation theorem is used as a topological input. It matches the geometric intuition that a closed surface has an inside and an outside.*

Theorem 7.4. *Every compact connected regular surface in* $\mathbb{R}^3$ *is orientable.*

Proof. Let Ω be the bounded component of $\mathbb{R}^3 \setminus S$. Near each point of S, the surface is a graph, and one can choose the unit normal pointing away from Ω. These local outward normals agree on overlaps and therefore define a global unit normal field. By the previous characterization, S is orientable. □

§8 A Geometric Definition of Area

Remark. *The area formula*

$$A(R) = \iint_D |\mathbf{x}_u \times \mathbf{x}_v| \, du \, dv$$

is not merely a formal definition. It is the limit of areas of small parallelograms approximating the image of small rectangles in the parameter domain.

Proposition 8.1. *Let $R = \mathbf{x}(D)$ be a region contained in a coordinate neighborhood, where $D \subset \mathbb{R}^2$ is a bounded domain. If D is divided into small rectangles and each image patch is approximated by the tangent parallelogram generated by $\mathbf{x}_u \Delta u$ and $\mathbf{x}_v \Delta v$, then the limiting sum of parallelogram areas is*

$$\iint_D |\mathbf{x}_u \times \mathbf{x}_v| \, du \, dv.$$

Remark. *Since*

$$|\mathbf{x}_u \times \mathbf{x}_v|^2 = |\mathbf{x}_u|^2 |\mathbf{x}_v|^2 - \langle \mathbf{x}_u, \mathbf{x}_v \rangle^2 = EG - F^2,$$

the geometric area element agrees with the expression obtained from the first fundamental form.

Appendix: A Brief Review of Continuity and Differentiability

Continuity

Definition 8.2. *A ball in $\mathbb{R}^n$ with center p and radius $\epsilon > 0$ is*

$$B_\epsilon(p) = \{q \in \mathbb{R}^n : |q - p| < \epsilon\}.$$

*A set $U \subset \mathbb{R}^n$ is **open** if every point of U is contained in some ball contained in U.*

Definition 8.3. *A map $F : U \subset \mathbb{R}^n \to \mathbb{R}^m$ is **continuous at** $p \in U$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that*

$$F(B_\delta(p)) \subset B_\epsilon(F(p)).$$

Proposition 8.4. *A map $F = (f_1, \dots, f_m) : U \subset \mathbb{R}^n \to \mathbb{R}^m$ is continuous if and only if each component function $f_i : U \to \mathbb{R}$ is continuous.*

Proposition 8.5. *The composition of continuous maps is continuous.*

Definition 8.6. *A continuous one-to-one map $F : A \subset \mathbb{R}^n \to \mathbb{R}^m$ is a **homeomorphism onto its image** if $F^{-1} : F(A) \to A$ is continuous.*

Theorem 8.7 (Intermediate Value Theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a)f(b) < 0$, then there exists $c \in (a, b)$ such that $f(c) = 0$.*

Theorem 8.8 (Extreme Value Theorem)**.** *A continuous real-valued function on a closed interval $[a, b]$ attains its maximum and minimum.*

Theorem 8.9 (Heine-Borel)**.** *Every open cover of a closed interval $[a, b]$ admits a finite subcover.*

Differentiability in $\mathbb{R}^n$

Definition 8.10. *In this book,* ***differentiable*** *means having continuous partial derivatives of all orders.*

Definition 8.11. *A map $F = (f_1, \ldots, f_m) : U \subset \mathbb{R}^n \to \mathbb{R}^m$ is differentiable if each component function f_i is differentiable.*

Definition 8.12. *Let $F : U \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable. The* ***differential*** *of F at p is the linear map*

$$dF_p : \mathbb{R}^n \to \mathbb{R}^m$$

defined by

$$dF_p(w) = (F \circ \alpha)'(0),$$

where $\alpha(0) = p$ and $\alpha'(0) = w$.

Remark. *In coordinates, if $F = (f_1, \ldots, f_m)$, then dF_p is represented by the Jacobian matrix*

$$\left(\frac{\partial f_i}{\partial x_j}(p) \right).$$

Theorem 8.13 (Chain Rule)**.** *If $F : U \subset \mathbb{R}^n \to \mathbb{R}^m$ and $G : V \subset \mathbb{R}^m \to \mathbb{R}^k$ are differentiable and $F(U) \subset V$, then*

$$d(G \circ F)_p = dG_{F(p)} \circ dF_p.$$

Theorem 8.14 (Inverse Function Theorem)**.** *Let $F : U \subset \mathbb{R}^n \to \mathbb{R}^n$ be differentiable. If dF_p is an isomorphism, then there are neighborhoods U_0 of p and V_0 of $F(p)$ such that $F : U_0 \to V_0$ is a diffeomorphism.*

Definition 8.15. *A value $a \in \mathbb{R}^m$ is a* ***regular value*** *of $F : U \subset \mathbb{R}^n \to \mathbb{R}^m$ if, for every $p \in F^{-1}(a)$, the differential dF_p is onto.*

Chapter III

The Geometry of the Gauss Map

Chapter IV

The Intrinsic Geometry of Surfaces

§1 Introduction

The intrinsic geometry of a surface is the part of its geometry which can be read from the first fundamental form alone. Thus notions such as length, angle, area, covariant derivative, geodesic curvature, and Gaussian curvature are intrinsic. The main point of this chapter is that many quantities which are defined by using the immersion in $\mathbb{R}^3$ actually depend only on the metric of the surface.

The central results are Gauss' Theorema Egregium, the theory of geodesics and parallel transport, and the Gauss-Bonnet theorem. Together they show how local curvature controls global topological information.

§2 Isometries; Conformal Maps

Definition 2.1. *Let $S, \bar{S}$ be regular surfaces. A diffeomorphism $\varphi : S \to \bar{S}$ is an **isometry** if, for every $p \in S$ and every $v, w \in T_pS$,*

$$\langle v, w\rangle_p = \langle d\varphi_p(v), d\varphi_p(w)\rangle_{\varphi(p)}.$$

*If this condition holds only in a neighborhood of each point, φ is called a **local isometry**.*

Proposition 2.2. *Let $x : U \to S$ and $\bar{x} : U \to \bar{S}$ be parametrizations with the same parameter domain. The map $\varphi = \bar{x} \circ x^{-1}$ is a local isometry if and only if the coefficients of the first fundamental forms agree:*

$$E = \bar{E}, \qquad F = \bar{F}, \qquad G = \bar{G}.$$

Remark. *Isometries preserve lengths of curves, angles between tangent vectors, and areas of regions. They need not preserve quantities depending on the second fundamental form. For example, a plane and a circular cylinder are locally isometric although their normal curvatures are different.*

Definition 2.3. *A diffeomorphism $\varphi : S \to \bar{S}$ is* ***conformal*** *if there is a positive differentiable function $\lambda : S \to \mathbb{R}$ such that*

$$\langle d\varphi_p(v), d\varphi_p(w)\rangle_{\varphi(p)} = \lambda(p)^2 \langle v, w\rangle_p$$

for all $p \in S$ and all $v, w \in T_pS$.

Proposition 2.4. *In local coordinates, $\varphi = \bar{x} \circ x^{-1}$ is conformal if and only if*

$$\bar{E} = \lambda^2 E, \qquad \bar{F} = \lambda^2 F, \qquad \bar{G} = \lambda^2 G$$

for some positive function λ.

Remark. *A conformal map preserves angles, but in general it changes lengths and areas by scale factors. Isometries are the special conformal maps with $\lambda = 1$.*

§3 The Gauss Theorem and the Equations of Compatibility

Let $x(u, v)$ be a parametrization of an oriented regular surface and let N be its unit normal. Write

$$E = \langle x_u, x_u\rangle, \qquad F = \langle x_u, x_v\rangle, \qquad G = \langle x_v, x_v\rangle$$

and

$$e = \langle x_{uu}, N\rangle, \qquad f = \langle x_{uv}, N\rangle, \qquad g = \langle x_{vv}, N\rangle.$$

Definition 3.1. *The* ***Christoffel symbols*** *Γ_{ij}^k are defined by the tangential parts of the second derivatives of x:*

$$\begin{aligned}
x_{uu} &= \Gamma_{11}^1 x_u + \Gamma_{11}^2 x_v + eN,\\
x_{uv} &= \Gamma_{12}^1 x_u + \Gamma_{12}^2 x_v + fN,\\
x_{vv} &= \Gamma_{22}^1 x_u + \Gamma_{22}^2 x_v + gN.
\end{aligned}$$

Remark. *The Christoffel symbols are determined only by E, F, G and their first derivatives. They are therefore intrinsic coefficients of the metric.*

Proposition 3.2. *The Weingarten equations may be written as*

$$\begin{aligned}
N_u &= a_{11} x_u + a_{21} x_v,\\
N_v &= a_{12} x_u + a_{22} x_v,
\end{aligned}$$

where the coefficients are determined by

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = - \begin{pmatrix} E & F \\ F & G \end{pmatrix}^{-1} \begin{pmatrix} e & f \\ f & g \end{pmatrix}.$$

Theorem 3.3 (Gauss Equation)**.** *The coefficients of the first and second fundamental forms satisfy*

$$K = \frac{eg - f^2}{EG - F^2}.$$

Moreover, the same function K can be expressed entirely in terms of E, F, G and their first and second derivatives.

Theorem 3.4 (Theorema Egregium)**.** *The Gaussian curvature of a regular surface is invariant under local isometries. Equivalently, K is determined by the first fundamental form.*

Remark. *Although the definition of K uses the Gauss map and the second fundamental form, Gauss' theorem says that K is an intrinsic quantity. This explains why a plane and a cylinder are locally isometric: both have $K = 0$.*

Theorem 3.5 (Compatibility Equations)**.** *The Gauss equation together with the Mainardi-Codazzi equations gives the compatibility conditions for the six functions*

$$E,\ F,\ G,\ e,\ f,\ g$$

to arise as the first and second fundamental forms of a surface.

Remark. *The fundamental theorem of local surface theory says that, under the compatibility equations and the positivity condition $EG - F^2 > 0$, these data determine a surface locally up to a rigid motion.*

§4 Parallel Transport. Geodesics

Definition 4.1. *Let $w(t)$ be a differentiable vector field along a curve $\alpha : I \to S$. The **covariant derivative** of w along α is*

$$\frac{Dw}{dt} = \left(\frac{dw}{dt}\right)^T,$$

the tangential component of the ordinary derivative in $\mathbb{R}^3$.

Remark. *If $\alpha(t) = x(u(t), v(t))$ and*

$$w(t) = a(t)x_u + b(t)x_v,$$

then

$$\frac{Dw}{dt} = Ax_u + Bx_v,$$

where

$$A = a' + \Gamma_{11}^1 au' + \Gamma_{12}^1(av' + bu') + \Gamma_{22}^1 bv',$$
$$B = b' + \Gamma_{11}^2 au' + \Gamma_{12}^2(av' + bu') + \Gamma_{22}^2 bv'.$$

Hence covariant differentiation depends only on the first fundamental form.

Definition 4.2. *A vector field w along α is **parallel** if*

$$\frac{Dw}{dt} = 0.$$

Proposition 4.3. *Given $w(t_0) \in T_{\alpha(t_0)}S$, there exists a unique parallel vector field along α with this initial value.*

Proposition 4.4. *Parallel transport preserves inner products. In particular, if v and w are parallel along α, then*

$$\frac{d}{dt}\langle v, w\rangle = 0.$$

Definition 4.5. *A parametrized curve $\alpha : I \to S$ is a **geodesic** if its tangent field is parallel:*

$$\frac{D\alpha'}{dt} = 0.$$

If α is parametrized by arc length, this is equivalent to saying that its acceleration is normal to the surface.

Remark. *In local coordinates $\alpha(t) = x(u(t), v(t))$, the geodesic equations are*

$$\begin{aligned} u'' + \Gamma^1_{11}(u')^2 + 2\Gamma^1_{12}u'v' + \Gamma^1_{22}(v')^2 &= 0, \\ v'' + \Gamma^2_{11}(u')^2 + 2\Gamma^2_{12}u'v' + \Gamma^2_{22}(v')^2 &= 0. \end{aligned}$$

Definition 4.6. *Let $\alpha(s)$ be a regular curve on an oriented surface, parametrized by arc length. Put*

$$t = \alpha'(s), \qquad n = N \times t.$$

*The **geodesic curvature** of α is*

$$k_g(s) = \left\langle \frac{Dt}{ds}, n \right\rangle.$$

Proposition 4.7. *A curve parametrized by arc length is a geodesic if and only if $k_g = 0$.*

Example 4.8. *The straight lines on a plane and the great circles on a sphere are geodesics. On a circular cylinder, the geodesics are the images of straight lines under the local isometry from the plane to the cylinder.*

Example 4.9 (Clairaut Relation)**.** *Let a surface of revolution be parametrized by*

$$x(u, v) = (f(v)\cos u, f(v)\sin u, g(v)).$$

If α is a geodesic and θ is the angle between α' and a meridian, then

$$f(v)\sin\theta = \textit{constant}.$$

§5 The Gauss-Bonnet Theorem and Its Applications

Definition 5.1. *A* ***regular region*** $R \subset S$ *is a compact connected region whose boundary is a finite union of piecewise regular simple closed curves.*

Definition 5.2. *Let* α *be a positively oriented piecewise regular curve and let* θ_i *be the exterior angle at its vertices. The sign of* θ_i *is chosen according to the orientation of the surface.*

Theorem 5.3 (Local Gauss-Bonnet Theorem)**.** *Let* R *be a simple region contained in a coordinate neighborhood of an oriented surface. Suppose that* ∂R *is a positively oriented piecewise regular curve, parametrized by arc length on each smooth arc. Then*

$$\sum_i \int_{\alpha_i} k_g \, ds + \sum_i \theta_i + \iint_R K \, d\sigma = 2\pi.$$

Corollary 5.4. *For a geodesic triangle with interior angles* $\varphi_1, \varphi_2, \varphi_3$*,*

$$\varphi_1 + \varphi_2 + \varphi_3 - \pi = \iint_R K \, d\sigma.$$

Remark. *On a surface of positive curvature, small geodesic triangles have angle sum greater than* π*; on a surface of negative curvature, the angle sum is less than* π*.*

Definition 5.5. *A* ***triangulation*** *of a regular region* R *is a finite decomposition of* R *into triangles whose intersections are either empty, a common vertex, or a common edge.*

Definition 5.6. *If a triangulation has* F *faces,* E *edges, and* V *vertices, the* ***Euler-Poincare characteristic*** *is*

$$\chi(R) = F - E + V.$$

Theorem 5.7 (Global Gauss-Bonnet Theorem)**.** *Let* R *be a regular region of an oriented surface and suppose that* ∂R *is piecewise regular and positively oriented. Then*

$$\sum_i \int_{\alpha_i} k_g \, ds + \sum_i \theta_i + \iint_R K \, d\sigma = 2\pi\chi(R).$$

Corollary 5.8. *If* S *is a compact oriented surface without boundary, then*

$$\iint_S K \, d\sigma = 2\pi\chi(S).$$

Remark. *The integral of the Gaussian curvature over a compact oriented surface is a topological invariant. This is the main bridge between local differential geometry and topology in this course.*

Example 5.9. *For the unit sphere,* $K = 1$ *and* $\text{area}(S^2) = 4\pi$*, hence*

$$\iint_{S^2} K \, d\sigma = 4\pi = 2\pi\chi(S^2).$$

Thus $\chi(S^2) = 2$.

Proposition 5.10. *A compact oriented surface with positive Gaussian curvature has positive Euler characteristic. In particular, a compact surface of genus* g *satisfies*

$$\chi = 2 - 2g.$$

Definition 5.11. *Let* v *be a vector field with an isolated singularity at* p*. The* ***index*** *of* v *at* p *is the winding number of the direction of* v *along a small positively oriented curve surrounding* p*.*

Theorem 5.12 (Poincare-Hopf Formula for Surfaces)**.** *If* S *is compact and oriented and* v *has only isolated singularities, then*

$$\sum_p I_p(v) = \chi(S).$$

§6 The Exponential Map. Geodesic Polar Coordinates

Definition 6.1. *Let* $p \in S$ *and* $v \in T_pS$*. Let* γ_v *be the geodesic with*

$$\gamma_v(0) = p, \qquad \gamma_v'(0) = v.$$

When $\gamma_v(1)$ *is defined, the* ***exponential map*** *at* p *is*

$$\exp_p(v) = \gamma_v(1).$$

Remark. *For small* v*, the exponential map is defined and differentiable. Moreover,*

$$d(\exp_p)_0 = \mathrm{id}_{T_pS}\,.$$

Hence $\exp_p$ *is a diffeomorphism from a neighborhood of* $0 \in T_pS$ *onto a neighborhood of* p *in* S*.*

Definition 6.2. *A neighborhood obtained as the image of a disk around* $0 \in T_pS$ *under* $\exp_p$ *is called a* ***normal neighborhood*** *of* p*.*

Proposition 6.3. *In a sufficiently small normal neighborhood, every point* q *is joined to* p *by a unique geodesic segment, namely the radial geodesic*

$$t \longmapsto \exp_p(tv), \qquad 0 \le t \le 1,$$

where $\exp_p(v) = q$*.*

Definition 6.4. *Choose polar coordinates* (ρ, θ) *in* T_pS*. The map*

$$x(\rho, \theta) = \exp_p(\rho e_\theta)$$

gives ***geodesic polar coordinates*** *on a punctured normal neighborhood.*

Proposition 6.5. *In geodesic polar coordinates the first fundamental form has the form*

$$d\rho^2 + G(\rho, \theta)d\theta^2.$$

Thus the radial curves $\theta =$ constant are unit speed geodesics and are orthogonal to the geodesic circles $\rho =$ constant.

Remark. *The behavior of $G(\rho, \theta)$ near $\rho = 0$ measures the spreading of geodesics issuing from p. Curvature controls this spreading: positive curvature makes nearby geodesics converge faster, while negative curvature makes them separate faster.*

Theorem 6.6 (Gauss Lemma)**.** *In a normal neighborhood of p, the radial direction is orthogonal to the directions tangent to the geodesic spheres centered at p.*

Proposition 6.7. *Let q be sufficiently close to p. Among all piecewise regular curves joining p to q inside a normal neighborhood, the radial geodesic segment has minimal length.*

Definition 6.8. *The* ***geodesic distance*** *between two points is*

$$d(p, q) = \inf L(\alpha),$$

where the infimum is taken over all piecewise regular curves joining p to q.

Remark. *Locally, the geodesic distance agrees with the length of the unique radial geodesic in a normal neighborhood.*

§7 Further Properties of Geodesics; Convex Neighborhoods

Theorem 7.1 (Existence and Uniqueness of Geodesics)**.** *Given $p \in S$ and $v \in T_pS$, there exists a unique geodesic $\gamma(t)$ defined for t in some interval around 0 such that*

$$\gamma(0) = p, \qquad \gamma'(0) = v.$$

Moreover, γ depends differentiably on the initial data (p, v).

Remark. *This follows from the local geodesic equations, which form a system of ordinary differential equations with differentiable coefficients.*

Definition 7.2. *A neighborhood $W \subset S$ is* ***convex*** *if any two points of W can be joined by a unique geodesic segment contained in W, and this segment depends differentiably on the two endpoints.*

Proposition 7.3. *Every point of a regular surface has a convex neighborhood.*

Proposition 7.4. *If $\alpha : [a, b] \to S$ is a minimizing curve parametrized by arc length, then α is a geodesic.*

Remark. *Thus geodesics are local candidates for shortest paths. The converse is only local: a geodesic segment need not remain minimizing after it is extended far enough.*

Definition 7.5. *A geodesic γ is **closed** if it is periodic as a parametrized curve. It is **simple closed** if its trace is a simple closed curve.*

Example 7.6. *Great circles on the sphere are closed geodesics. On a circular cylinder, the closed geodesics are the parallels obtained from horizontal lines in the plane. Helices are also geodesics, but they are not closed.*

Remark. *Convex neighborhoods are used to compare local and global behavior. They allow one to replace short pieces of arbitrary curves by unique minimizing geodesic segments.*

Appendix: Proofs of the Fundamental Theorems of the Local Theory of Curves and Surfaces

Remark. *The fundamental theorem for curves is proved by solving the Frenet system. Given functions $\kappa > 0$ and τ, one first constructs an orthonormal moving frame satisfying the Frenet equations and then obtains the curve by integrating its tangent vector. Uniqueness follows from uniqueness for ordinary differential equations and from the fact that two initial orthonormal frames differ by a rigid motion.*

Remark. *The fundamental theorem for surfaces is analogous but technically richer. The prescribed functions E, F, G, e, f, g must satisfy the Gauss and Mainardi-Codazzi equations. These are precisely the integrability conditions for the first-order system which reconstructs the frame x_u, x_v, N. Once the frame is recovered, the surface is obtained by integrating x_u and x_v.*

Theorem 7.7 (Fundamental Theorem of Local Surface Theory)**.** *Let E, F, G, e, f, g be differentiable functions on a domain $U \subset \mathbb{R}^2$ with*

$$E > 0, \qquad EG - F^2 > 0.$$

If the Gauss and Mainardi-Codazzi equations hold, then for each point of U there is a neighborhood on which these functions are the coefficients of the first and second fundamental forms of a regular surface. The surface is unique up to a rigid motion of $\mathbb{R}^3$.

Chapter V

Global Differential Geometry

Bibliography and Comments

Hints and Answers

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